Revised Agricultural Field Scale Irrigation Requirement Simulation

Version 1.2

User Guide

September 2018

Update History

Date	People	Version	Description
2018-06-18	Andre	1.0	Initial commit
2018-09-23	Meng	1.1	1. Add section for user registration 2. Add description for using predefined soil type from soil DB 3. Change the description for using Soil Map tools since the back-end DB for soil data has been implemented. 4. Adjust the order of sections for creating water use permit to make whole process more logically.
2018-11-06	Meng	1.2	1. Change the description for creating polygon on Soil Map tool since new feature for editing polygon is implemented. 2. Add update history for future references.

Introduction

The **Agricultural Field Scale Irrigation Requirements Simulation** (AFSIRS) model was developed in the 1980's by Allan Smajstrla, with the University of Florida (UF) Department of Agricultural and Biological Engineering (ABE). The AFSIRS was initially described as a numerical simulation model which allows the user to estimate **irrigation requirements** (IRR) for Florida crops, soils, irrigation systems, growing seasons, climate conditions and irrigation management practices. IRR for crop production is the amount of water, exclusive of precipitation, that must be applied to meet a crop's **evapotranspiration** (ET) requirements without significant reduction in yield. IRR, as defined in AFSIRS, does not include leaching, freeze protection, or crop cooling requirements, even though water for these purposes may be applied through an irrigation system.

The AFSIRS model is based on a water budget of the crop root zone and the concept that crop ET can be estimated from **potential evapotranspiration** (ETo) and **crop water use coefficients** (Kc). The water budget includes inputs to the crop root zone from rain and irrigation, and losses from the root zone by drainage and ET. The **water storage capacity** in the crop root zone is defined as the multiple of the **water-holding capacity** of the soil and the **depth of the effective root zone** for the crop being grown.

Current AFSIRS Modifications

In 2014, the Florida Department of Agriculture and Consumer Services (FDACS), Office of Agricultural Water Policy (OAWP) requested a group of UF researchers, Institute of Food and Agricultural Sciences (IFAS), to develop a project to modernize AFIRS in terms of computer science and water balance technologies. The version of AFSIRS described in the following user guild has been rewritten from the original FORTRAN into Java, databases developed for soil, weather, irrigation efficiency, and crop coefficients, and the addition of a global information system (GIS) system to determine site specific soil characteristics.

User Guide

The improved interface of the rewritten AFSIRS model is user-friendly compared to the previous FORTRAM model. However, the same information is required as model input. The soil characteristics for multiple soils in a given field are provided using a GIS tool, or you could built up your own soil characteristics by using the system-defined soil characteristics data. Output of irrigation requirement from the model is provided in the same text file used in the previous version but also in graphical form. These new interface features are illustrated below.

I. Register your account

The AFSIRS online require all the user have their own account to create/save the data set and run the simulation. On the home page (https://afsirs-online-test.herokuapp.com/) as below, Click the “Register” on the page (Fig. 1).

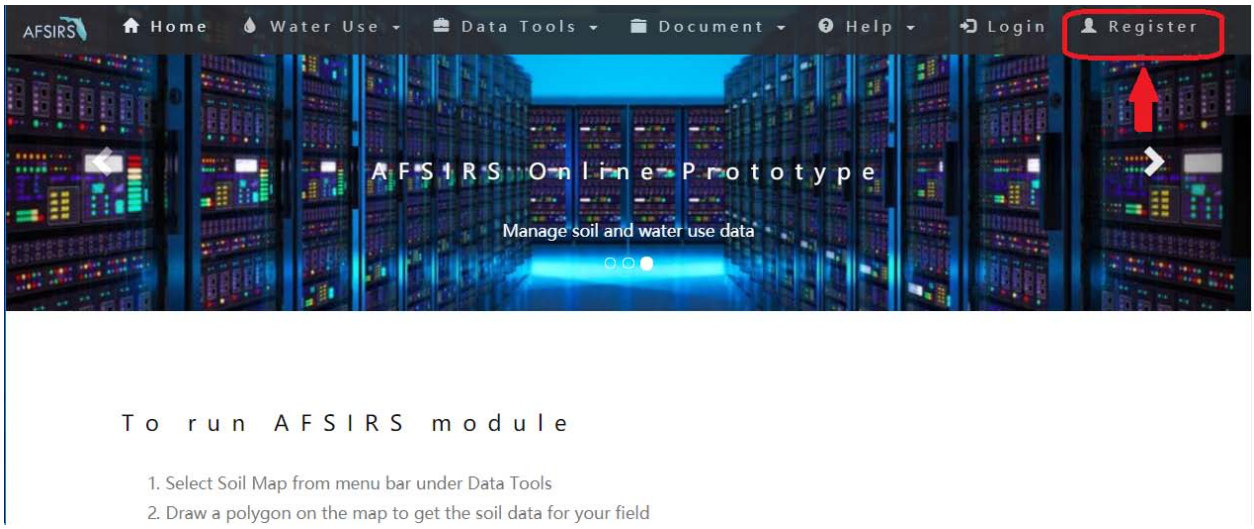

Figure 1.

On the register page(Fig. 2), please input your user name and password. By clicking Register button, the system will create your account. Your password will be stored with encryption for security purpose.

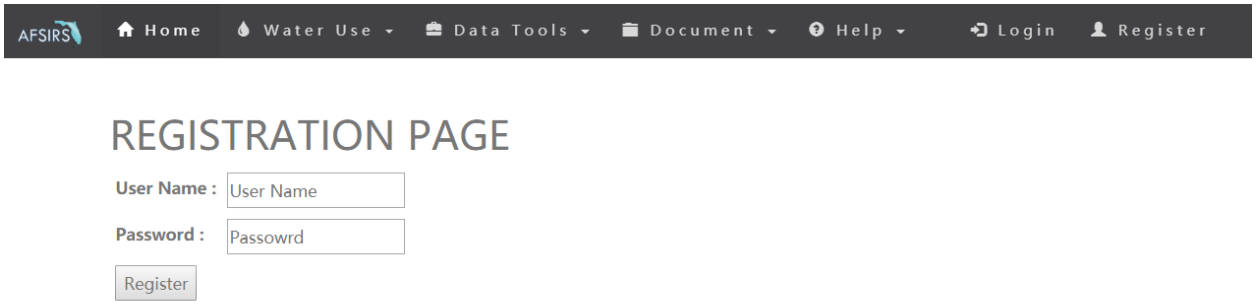

Figure 2.

II. Create the soil data set

There two ways to create the data set in the system, one is GIS tool, which is the new feature for our AFSIRS online compared to the old Fortran version; another way is to use pre-defined soil type data from the soil data base to create your own soil profile data.

This section will describe about how to use GIS tool to locate your field and create soil data set for running AFSIRS model. If you have already had your data ready online or in a external JSON file from previous version of AFSIRS desktop, or if you are going to use pre-defined soil type data from soil data base, you can skip this section and go to the next one (Create the simulation data set).

In the AFSIRS home page, select “Data Tools” from menu bar and choose the “Soil Map” (Fig. 3) to enter the GIS tool page.

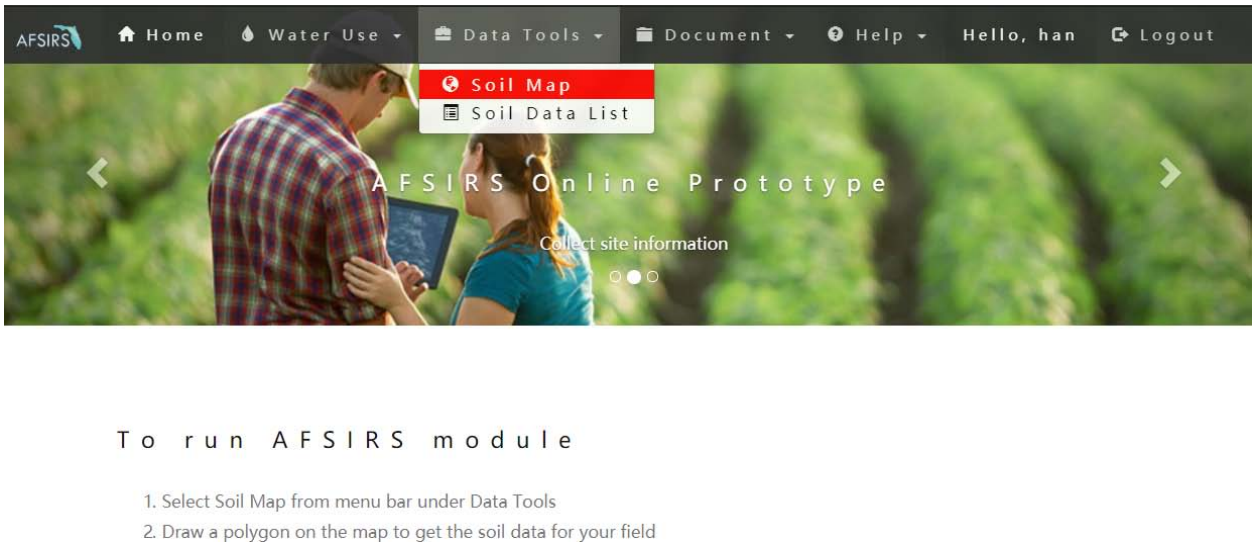

Figure 3.

A new window (Fig. 4) will show up with an GIS map and other components as below,

1. Primary control buttons
2. Data set information
3. Soil data table (only show up after drawing the polygon)
4. GIS Map
5. Quick Guide

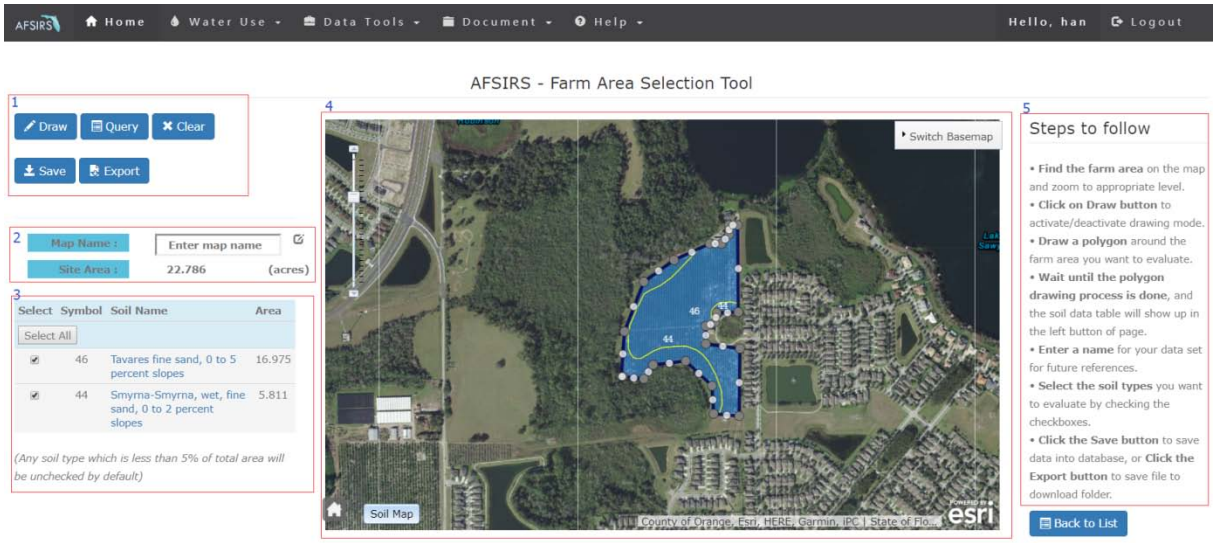

Figure 4.

This GIS map supports the following GIS feature to help locate your farm field,

- Holding left-click of mouse to drag and change the current map location,
- Using mouse scroll to zoom in/out
- Clicking “switch Basemap” button on the right-top to change to other base-map for your convenience (Fig. 5-1)
- To edit the polygon you have already created, you can drag and drop the vertices to change the shape of polygon. Once it is done, a new query will start automatically and update the soil table with new query result. (Fig. 5-2)
- You can also move the whole polygon to a new location by clicking the center of polygon, then drag and drop it to where you would like. Similarly a new query will be done after you finish the editing. (Fig. 5-3)

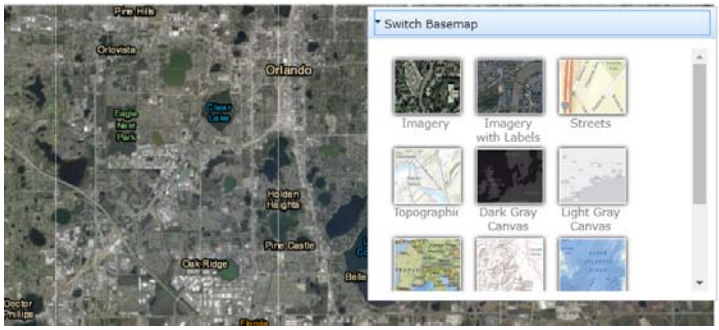

Figure 5-1.

ymbol	Soil Name	Area
20	Immokalee fine sand	1.207
34	Pomello fine sand, 0 to 5 percent slopes	0.951
54	Zolfo fine sand, 0 to 2 percent slopes	0.350

pe which is less than 5% of total area will be

Symbol	Soil Name	Area
20	Immokalee fine sand	1.530
34	Pomello fine sand, 0 to 5 percent slopes	0.951
54	Zolfo fine sand, 0 to 2 percent slopes	0.350
41	Samsula-Hontoon-Basinger association, depressional	0.154

Figure 5-2.

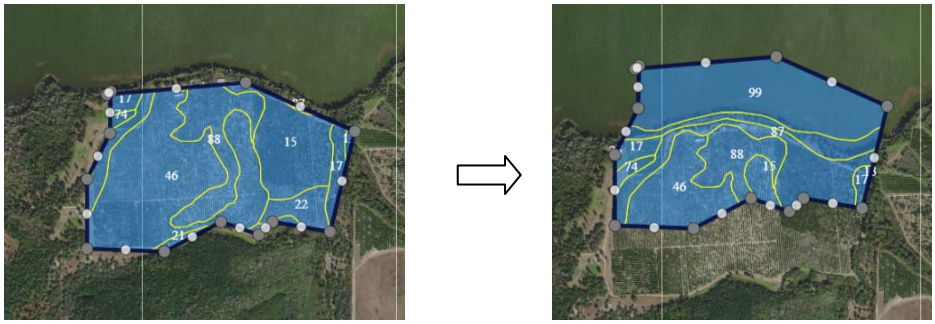

Figure 5-3.

Quick guide

On the page (Fig. 4), you can find a quick guide on the right side of the page. It shows how to use this tool. You can easily follow those steps to create your data set.

Primary buttons

On the left-top of the page, you can find the primary buttons of the tool. Their functionality descriptions are as below,

- Draw – It actves/deactives the drawing mode which allow you using left-clicking on the map to create vertex of your polygon. A double left-clicking will tell tool to end your drawing.
- Query – It will query the soil data within the drawn polygon area. Click it when you want to refresh your soil data table on the left-buttom. It will be disabled during the data transmission process.
- Clear – It will clear the current polygon and query result.
- Save – It will save your current polygon into online database for future simulation usage. You can retrieve your saved records from a list page which could be accessed under Data tools menu.
- Export – It will export the current polygon data into a *.JSON file and download it into your local drive.

Soil data table

After query process is done, this data table will show up with a list of soil series which are included inside the polygon. The number on the sub-polygon is matched with symbol column in the table. By clicking the name of the soil series, a popup will come out to show the detail component of this soil profile (Fig. 6). By default, the small sub-polygon will be unchecked in order to accelerate the simulation speed.

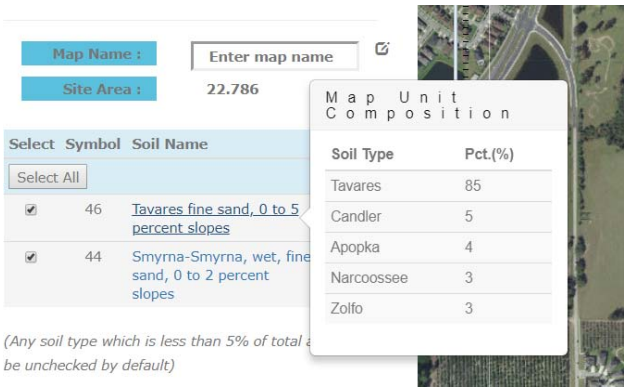

Figure 6

III. Create the simulation data set (Water-Use Permit)

Once you have your soil data ready for the simulation, you can go to create your simulation data set (a.k.a. water-use permit) by selecting “Water Use” from the menu bar and choose the “Create Permit” (Figure 7).

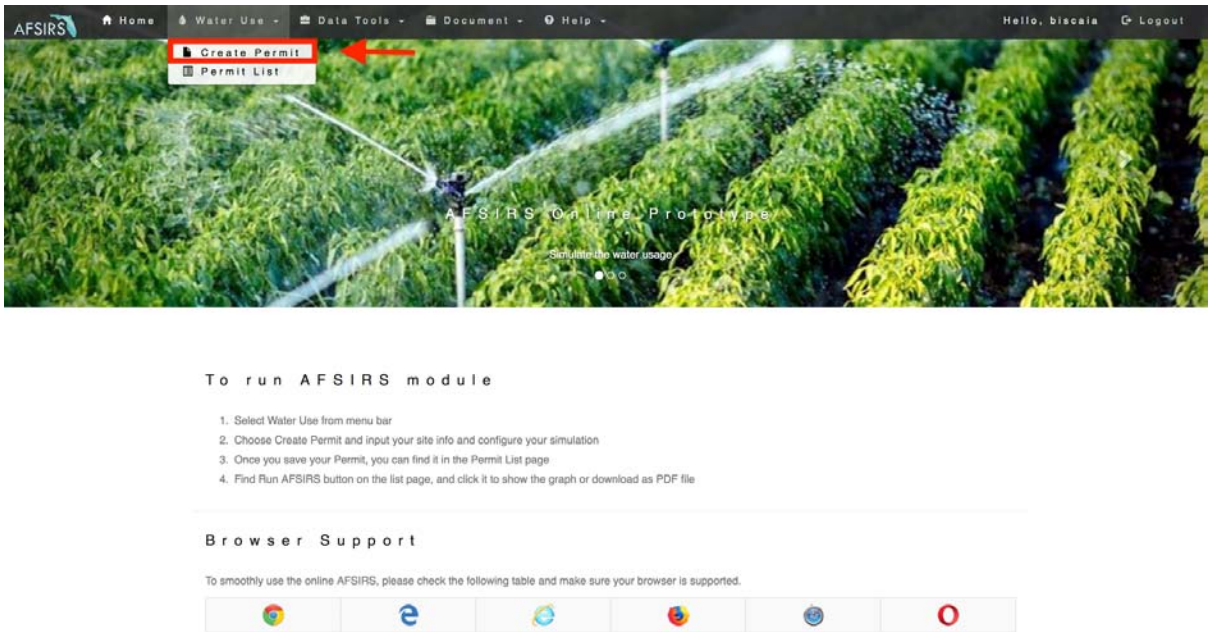

Figure 7.

Creating a water permit

General

In the “General” section of “Create Permit” (Fig. 7), users will provide information regarding personal information and crop grown:

1) *Permit ID:*
2) *Owner name:*
3) *Crop cycle and crop type:* to identify the crop cycle, which can be perennial or annual, and crop cultivated.

4) *Simulation start date and end date (e.g., month and day):* Start and end dates will define the duration of the irrigation season. This is the period of each year for which the water budget and irrigation requirements will be calculated. If January 1 (1/1) is specified as the beginning of the irrigation season and December 31 (12/31) is specified as the end of the season, then the irrigation season will be defined as the entire year and IRR will be calculated for that period, that is, assuming that irrigation is practiced throughout the year. The irrigation season may be defined to extend from one calendar year to the next. Thus, if the beginning of the irrigation season is defined as November 1 and the end of the season as March 1, IRR will be calculated for that period, beginning with the November 1 data from the first year of the climate record, and extending through March 1 of the last year of the climate record at the site indicated by the user later.

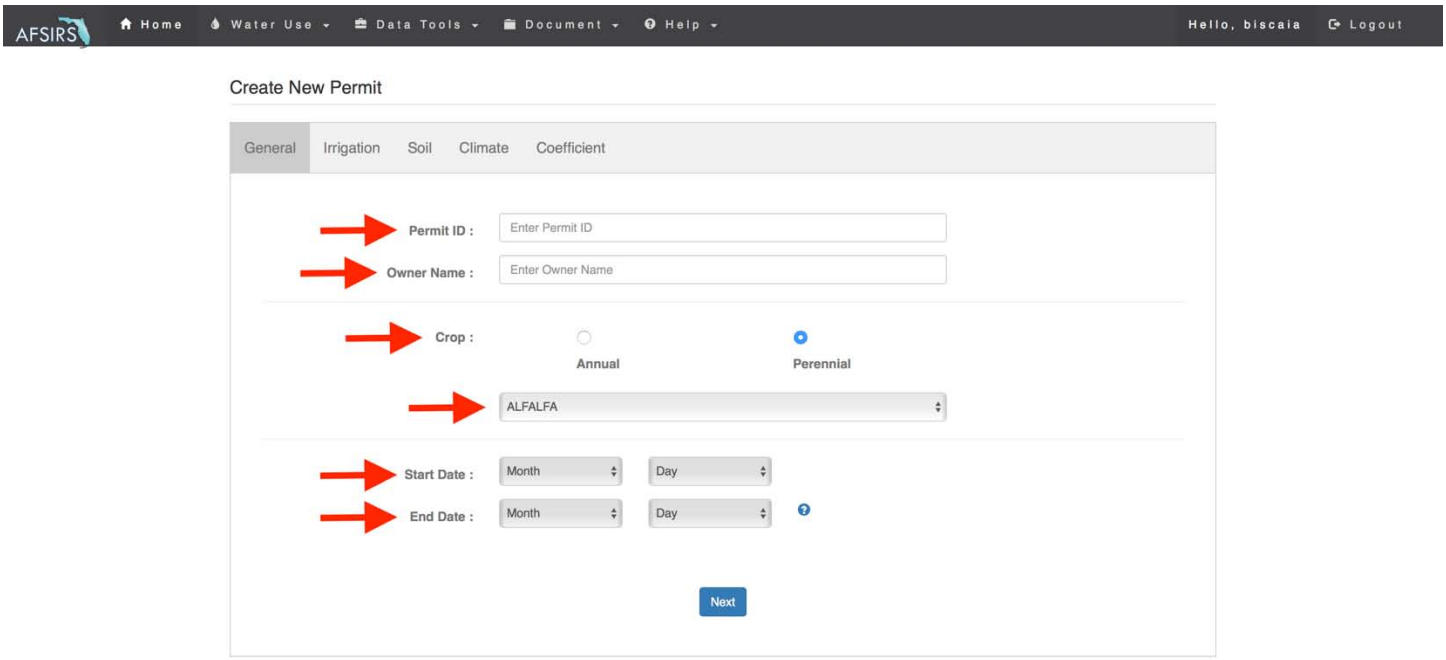

Figure 8. General section information to create a new permit.

Irrigation management

In the AFSIRS model, the net irrigation requirement (NIR) is calculated as the depth of water required to restore the soil water to field capacity (maximum soil water storage capacity) in the irrigated crop root zone. The gross irrigation requirement (IRR) is calculated as NIR divided by the irrigation system application efficiency (EFF), where EFF is expressed as a decimal fraction. IRR is always greater than NIR, and IRR is the irrigation requirement of the production system, including the effectiveness of water application with the irrigation system used. Therefore, in the "Irrigation" subsection of the "Create Permit" (Fig. 9), users need to provide irrigation information regarding:

1) *Irrigation type:* to identify the irrigation system to be used. Irrigation systems can be trickle drip, trickle spray, multiple sprinkler, sprinkler, crown flood (Citrus) and flood (Rice).
2) *Calculation type*: to select the calculation of net or gross irrigation requirement.
3) *Depth of water application:* to select how irrigation systems will be managed (e.g., irrigate to field capacity, a fixed water depth per application, or a deficit irrigation application).
4) *Irrigation coefficients:* default irrigation coefficients are automatically estimated according to irrigation information aforementioned. However, AFSIRS users might edited such coefficient as needed. Irrigation coefficients are system efficiency, surface fraction, ET fraction and water table depth.

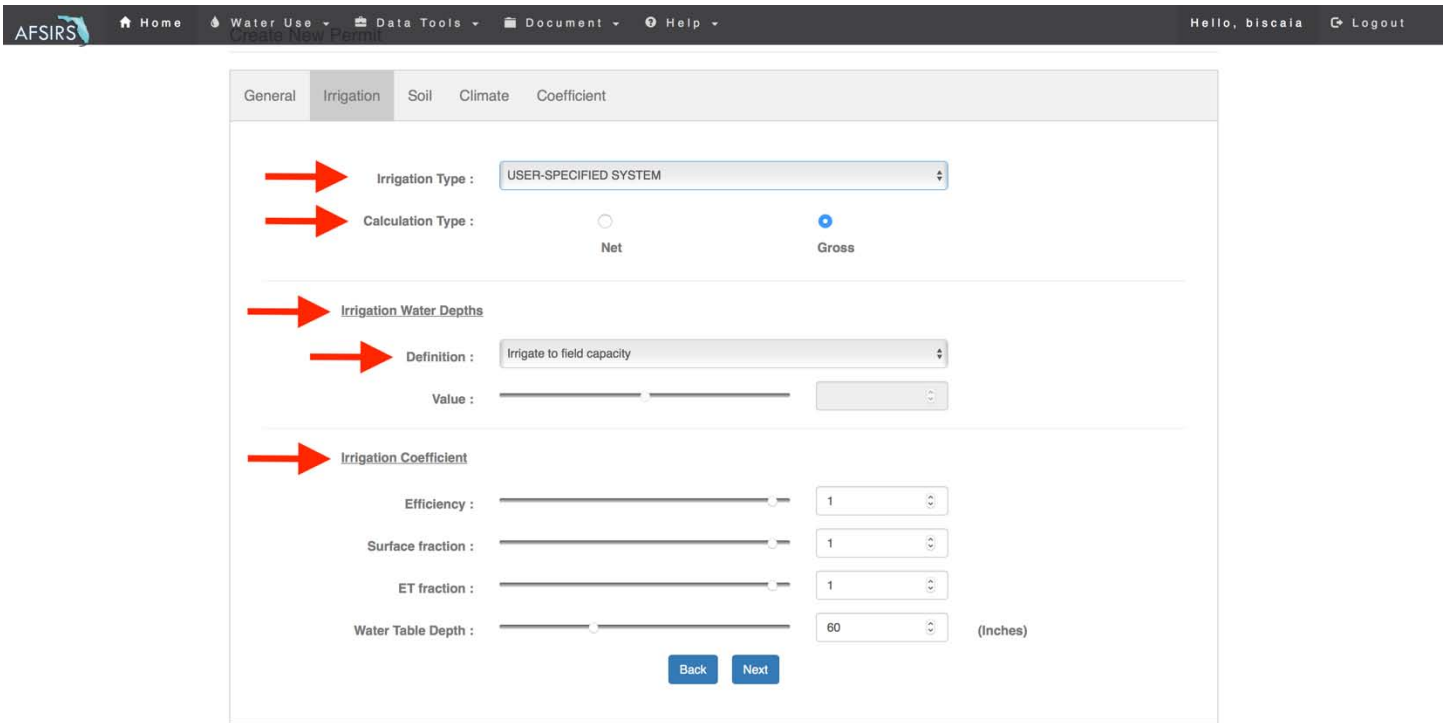

Figure 9.

Soil Characteristics

The daily water budget used in the AFSIRS model simulates the complex dynamic processes of soil water infiltration, redistribution, and extraction by evapotranspiration (ET) as steady state processes on a daily basis. This level of simulation model development was required to produce a functional model that could address the wide variety of crops, soils, and irrigation systems typical of Florida, and to estimate IRR in a reasonable amount of computer time. This simulation time scale is adequate to permit the interactions of distributions of rainfall and ET on IRR to be calculated. In addition, the AFSIRS model takes into consideration the available soil water, which is defined as the water stored between field capacity and the permanent wilting point (PWP). PWP is the soil water potential beyond which a crop cannot extract water and dies. Because of the naturally-occurring range in soil water-holding capacities within a soil series, the

soils data base lists both maximum and minimum expected values. Thus, the AFSIRS model allows the user to select the maximum, minimum, or average (arithmetic average of the maximum and minimum) values during each simulation.

In the “Soil” subsection of the “Create Permit” section, the new AFSIRS interface uses the map function of the Geographical Information System interface (GIS) to identify the soil characteristics or use pre-defined soil type to let you create your own soil profile.

For GIS way, by following the previous section, you should already create now your soil data record in the system. Now you can select it from the drop down list on the page (Fig. 10).

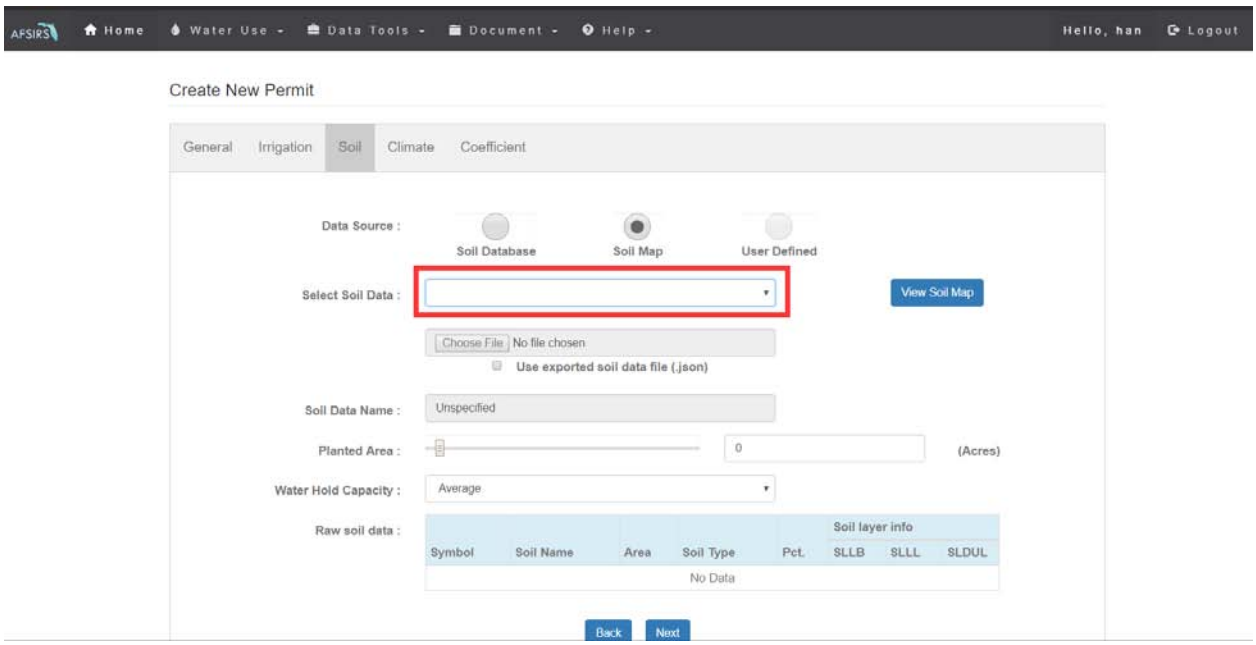

Figure 10.

Once you choose your record, the data should be loaded into the table below (Fig. 11).

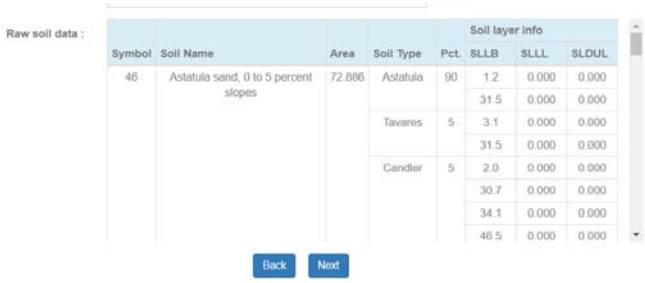

Figure 11.

Soil characteristics are “Planted area”, “Soil type”, soil layers (SLLB), soil permanent wilting point (SLLL) and soil field capacity (SLDUL).

For pre-defined soil data way, you should switch the data source to “Soil Database” as below (Fig. 12), and the pre-defined soil type list will show up.

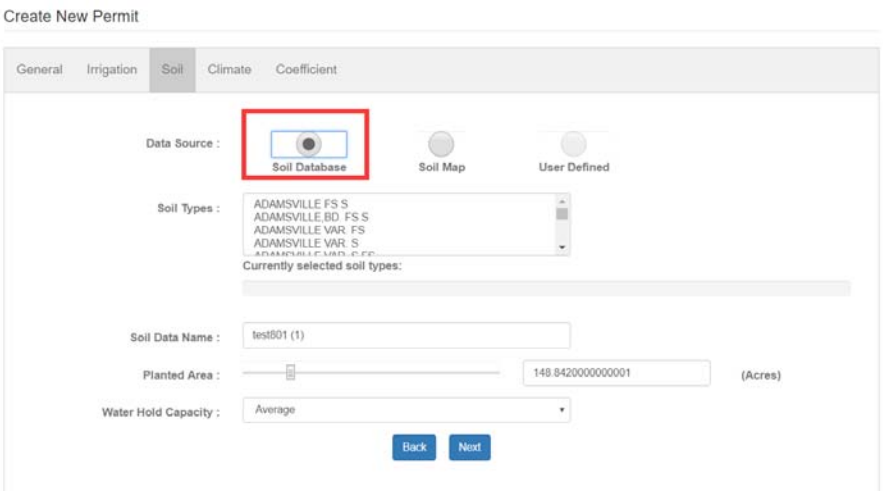

Figure 12.

The list supports multiple-selection. To do that, please hold the ctrl key while left-click the items on the list. Once you select soil types on the list, a proportion information of each soil type will show up (Fig. 13). By default, each soil type will have same proportion in your soil profile. You can change that by drag the bar or input your number directly.

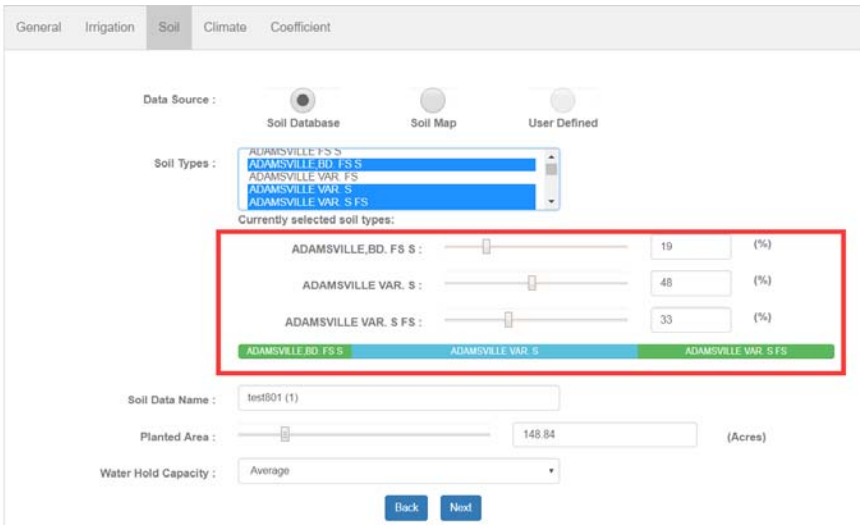

Figure 13.

Climate Information

The AFSIRS model uses historical climate data to simulate daily IRR for crop seasons for the number of years of climate record. This long-term record of IRR is then used to calculate statistical parameters including the probabilities of occurrence of IRR for various portions of the crop growing season. These extreme values may be required for irrigation system design or irrigation water use permitting.

Once the .JSON file was previously used to select the area location in the "Soil" subsection of the "Create Permit", the "Climate" subsection will automatically identify the nearest weather station location from the Florida Automated Weather Network (FAWN) and estimate the historical evapotranspiration and rainfall for that location (Fig. 14). However, the user still can choose any weather station from the list by unselecting the check box.

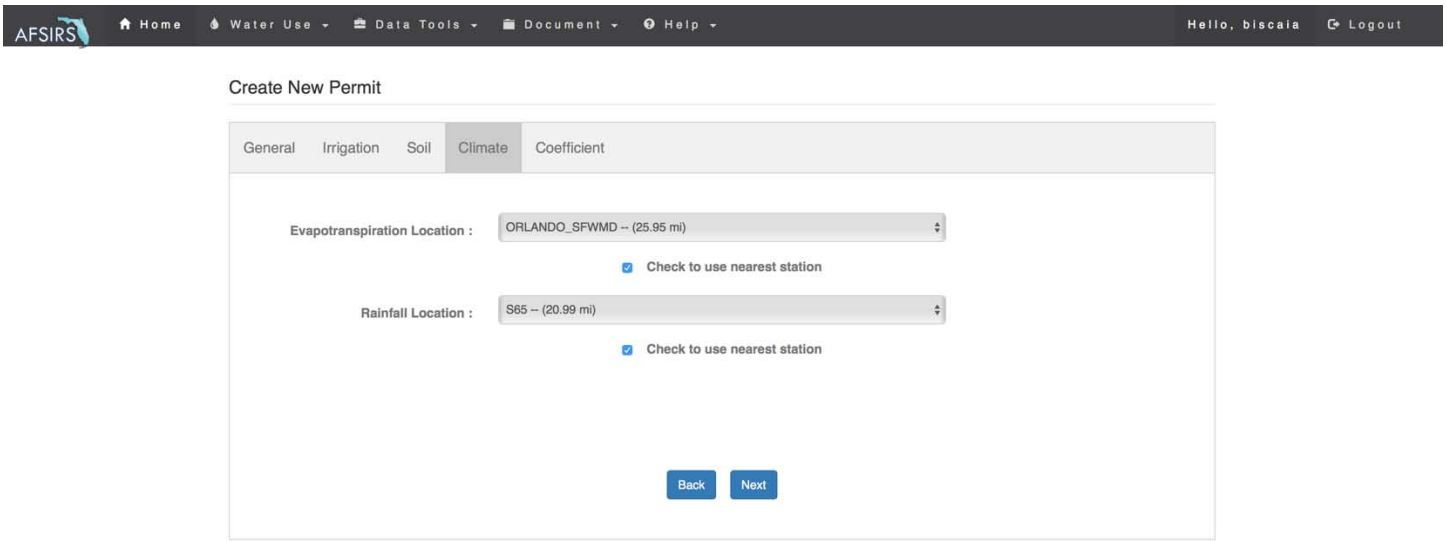

Figure 14.

Coefficients Information

The last subsection of the “Create Permit” section is the “Coefficient”. In this subsection default coefficients for “Irrigated Root Zone Depth”, “Crop water use coefficients”, “Fraction of Growing Season”, and “Allowable Soil Water Depletions” are automatically set up according to the irrigation system, water table depth and crop selected in the previous subsections of the “Create Permit” (Fig. 15). However, users can manually edit such coefficients as needed by unselecting the “Default” box from the “Crop data” of the “Coefficients” subsection of the “Create Permit”.

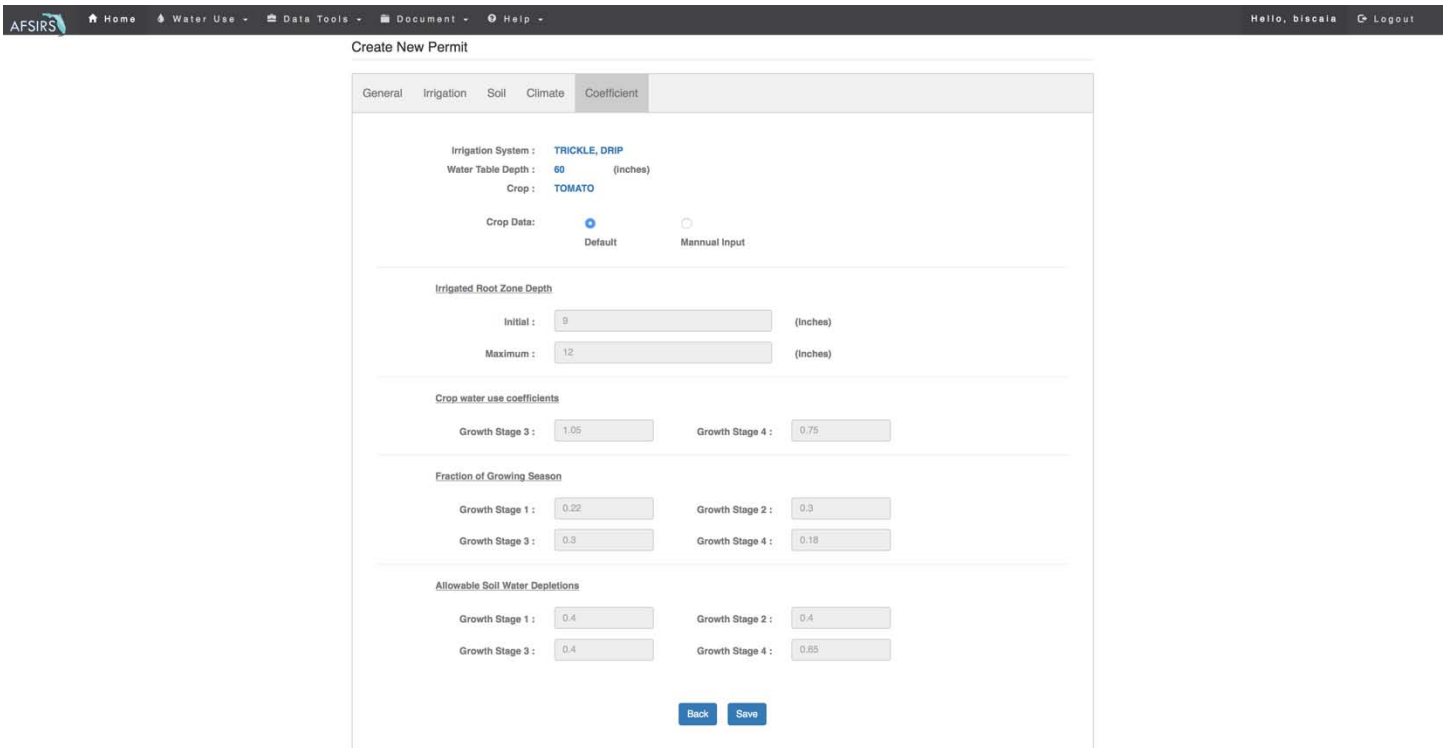

Figure 15.

After this section, the user should click on save, creating a permit based on inputs just provided in previous steps. This will redirect the page for the “Permit List” section of the “Water use”, where users might run the AFSIRS model (Fig. 16).

Figure 16.

Outputs

Model output includes the model-generated values of the length of growing season (days), length of climate record for the climate data base used, irrigation system design efficiency, fraction of the soil surface assumed to be irrigated with the irrigation system specified, soil water-holding capacity for the soil type specified, and tables of the simulated IRR and statistical characteristics.

After running the model, an interactive graph will open (Fig. 17). In this graph, user can selected the results of irrigation requirement, rainfall, crop evapotranspiration, or all of them for each type of soil identified in the soil characteristic subsection, and an weighted average.

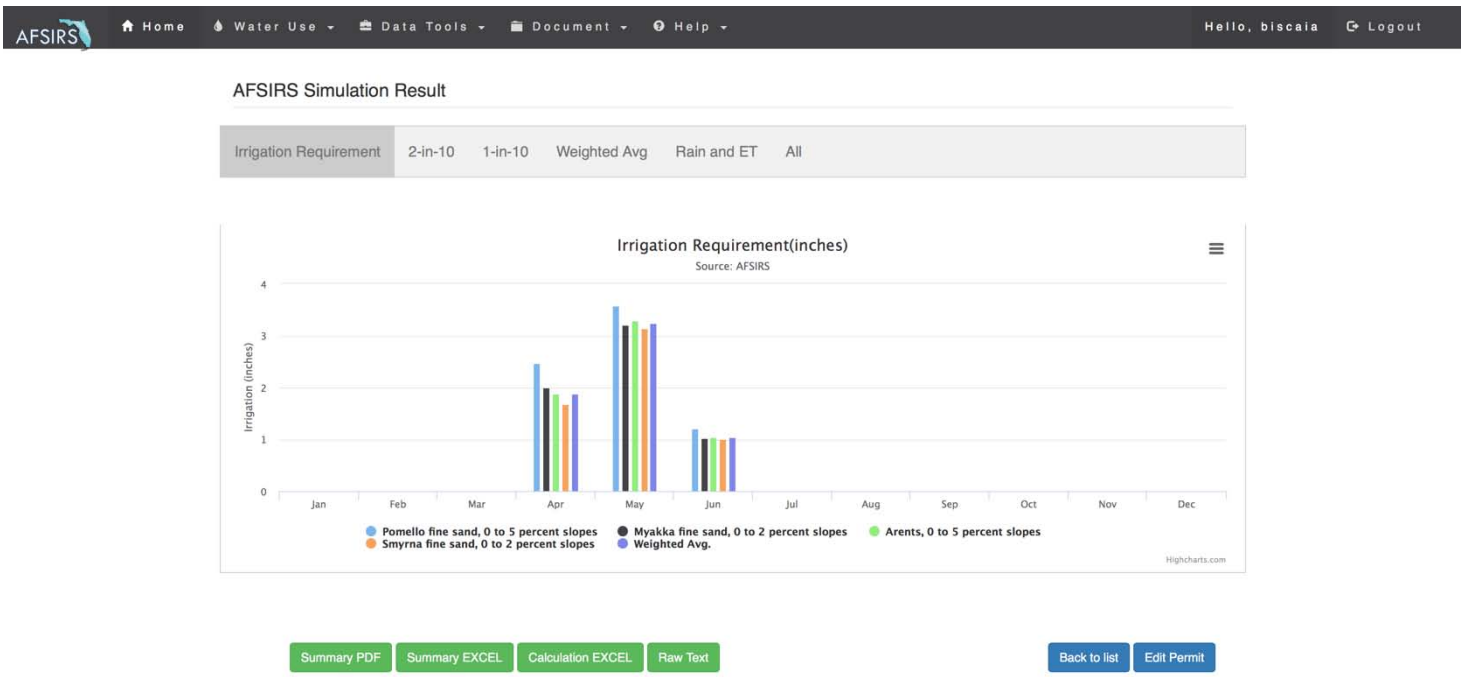

Figure 17.

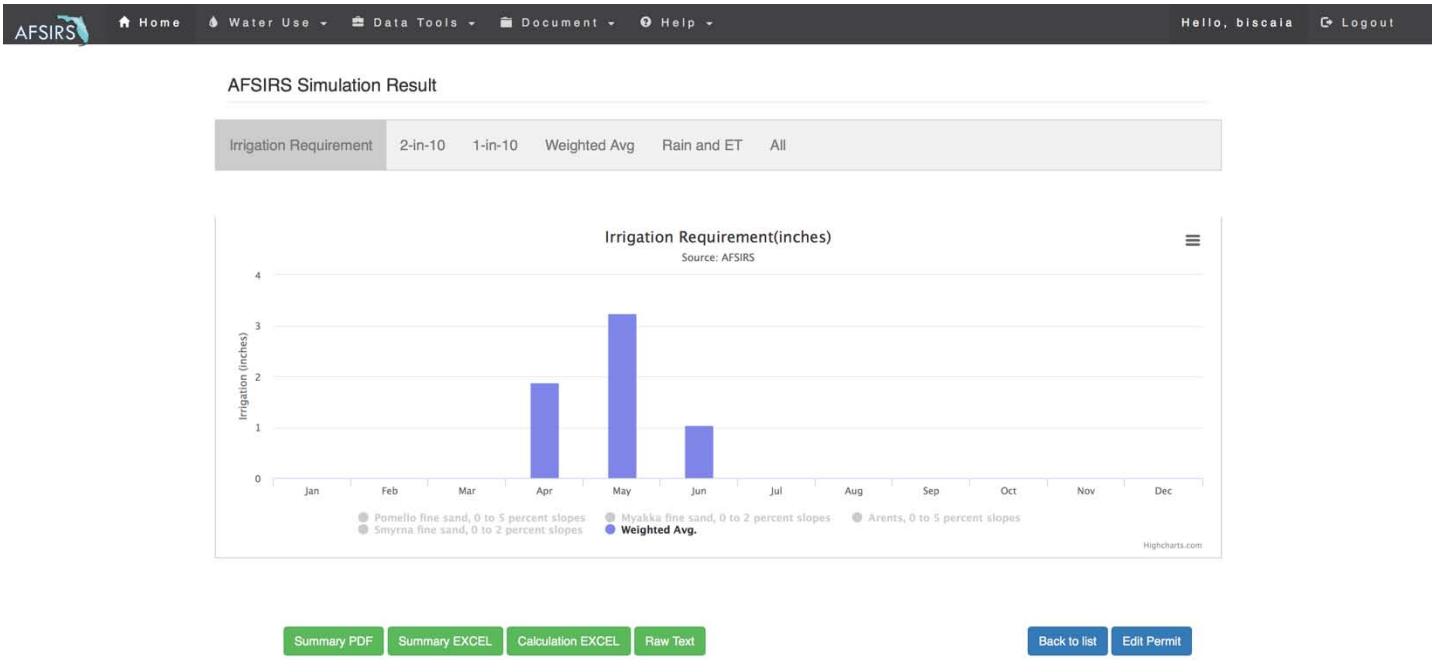

Figure 18.

In addition to graphs, there is a summary PDF that can be required by the user. In this file, it will be provided a seasonal and monthly IRR tables, and a soil characteristic table. The first table, called “Weather data during the season”, is monthly averaging rainfall, reference evapotranspiration (ET) and crop evapotranspiration (ET crop) for the entire growing season (for annual crops) or for the entire year (for perennial crops). Similarly, the second table list summarize the total irrigation requirement for all types of soils per month. Next list of tables is the summary of the total water requirement by soil type. While, the last table, called “Soil data”, will provide all information used by the model for each soil before identified by the user.

AFSIRS MODEL: INTERACTIVE VERSION 6.3.2-SNAPSHOT

Owner Name : **John Smith**
Crop : **TOMATO**
Crop Simulation End Date : **6/30**
Site Area (ACRES):**150.434**
Irrigation Option: **NET**
Selected Soil Area :**150.15**
KC :**1.05 0.75**

Permit ID : **ID 0001**
Irrigation Method : **TRICKLE, DRIP**
Period of Record : **1965-2015**
ET Station : **ORLANDO_SFWMD**
Irrigation efficiency : **1.0**
Depth of Water Table :**60.0**

Map Name : **Unspecified-2**
Crop Simulation Start Date : **4/1**
Planted Area (ACRES): **150.149**
Rainfall Station : **S65**
Soil Surface Irrigated : **0.5**
Water Holding Capacity :**Average**

Weather data during the season:

Inches/Month													
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total
Mean Rainfall	0.00	0.00	0.00	2.12	4.05	8.17	0.00	0.00	0.00	0.00	0.00	0.00	14.34
Mean ET	0.00	0.00	0.00	5.95	6.68	5.75	0.00	0.00	0.00	0.00	0.00	0.00	18.39
Mean ET (Crop)	0.00	0.00	0.00	4.20	6.37	5.67	0.00	0.00	0.00	0.00	0.00	0.00	16.24

Total Irrigation requirements (all soil series):

Irrigation Weighted Average (Inches)													
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total
Mean Irr Req	0.00	0.00	0.00	1.89	3.24	1.04	0.00	0.00	0.00	0.00	0.00	0.00	6.17
2-In-10 Irr Req	0.00	0.00	0.00	2.35	4.05	1.30	0.00	0.00	0.00	0.00	0.00	0.00	7.71
1-In-10 Irr Req	0.00	0.00	0.00	2.59	4.46	1.44	0.00	0.00	0.00	0.00	0.00	0.00	8.48
Irrigation Weighted Average (Million Gallons)													
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total
Mean Irr Req	0.00	0.00	0.00	7.69	13.21	4.26	0.00	0.00	0.00	0.00	0.00	0.00	25.16
2-In-10 Irr Req	0.00	0.00	0.00	9.60	16.51	5.32	0.00	0.00	0.00	0.00	0.00	0.00	31.43
1-In-10 Irr Req	0.00	0.00	0.00	10.55	18.18	5.86	0.00	0.00	0.00	0.00	0.00	0.00	34.59

Deviations List:

Deviations

Irrigation system has been changed to :**TRICKLE, DRIP**
Irrigation Requirement has been changed to :**NET**
Irrigation application efficiency has been changed to :**1.0**
Planted area has been changed to :**150.149**

Total Irrigaiton Requirements by soil series:

Soil Series Name : **Pomello fine sand, 0 to 5 percent slopes**

Soil Map Unit Symbol# : **34** Soil Map Unit Code : **1483874** Soil Percentage : **16.92**

Details in Inches													
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total
Mean Irr Req	0.00	0.00	0.00	2.47	3.58	1.20	0.00	0.00	0.00	0.00	0.00	0.00	7.25
2-in-10 Irr Req	NA	NA	NA	2.98	4.32	1.45	NA	NA	NA	NA	NA	NA	8.75
1-in-10 Irr Req	NA	NA	NA	3.22	4.67	1.57	NA	NA	NA	NA	NA	NA	9.46
Details in Million Gallons													
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total
Mean Irr Req	0.00	0.00	0.00	10.06	14.58	4.91	0.00	0.00	0.00	0.00	0.00	0.00	29.55
2-in-10 Irr Req	0.00	0.00	0.00	12.14	17.60	5.92	0.00	0.00	0.00	0.00	0.00	0.00	35.67
1-in-10 Irr Req	0.00	0.00	0.00	13.13	19.03	6.41	0.00	0.00	0.00	0.00	0.00	0.00	38.56
Details in Inches(Weighted)													
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total
Mean Irr Req	0.00	0.00	0.00	0.42	0.61	0.20	0.00	0.00	0.00	0.00	0.00	0.00	1.23
2-in-10 Irr Req	NA	NA	NA	0.50	0.73	0.25	NA	NA	NA	NA	NA	NA	1.48
1-in-10 Irr Req	NA	NA	NA	0.54	0.79	0.27	NA	NA	NA	NA	NA	NA	1.60
Details in Million Gallons(Weighted)													
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total
Mean Irr Req	0.00	0.00	0.00	1.70	2.47	0.83	0.00	0.00	0.00	0.00	0.00	0.00	5.00
2-in-10 Irr Req	0.00	0.00	0.00	2.05	2.98	1.00	0.00	0.00	0.00	0.00	0.00	0.00	6.03
1-in-10 Irr Req	0.00	0.00	0.00	2.22	3.22	1.08	0.00	0.00	0.00	0.00	0.00	0.00	6.52

Soil Series Name : **Myakka fine sand, 0 to 2 percent slopes**

Soil Map Unit Symbol# : **22** Soil Map Unit Code : **1483862** Soil Percentage : **18.89**

Details in Inches													
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total
Mean Irr Req	0.00	0.00	0.00	2.00	3.20	1.02	0.00	0.00	0.00	0.00	0.00	0.00	6.23
2-in-10 Irr Req	NA	NA	NA	2.50	4.00	1.27	NA	NA	NA	NA	NA	NA	7.77
1-in-10 Irr Req	NA	NA	NA	2.75	4.40	1.40	NA	NA	NA	NA	NA	NA	8.55
Details in Million Gallons													
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total
Mean Irr Req	0.00	0.00	0.00	8.16	13.06	4.16	0.00	0.00	0.00	0.00	0.00	0.00	25.38
2-in-10 Irr Req	0.00	0.00	0.00	10.19	16.30	5.20	0.00	0.00	0.00	0.00	0.00	0.00	31.68
1-in-10 Irr Req	0.00	0.00	0.00	11.21	17.93	5.72	0.00	0.00	0.00	0.00	0.00	0.00	34.85
Details in Inches(Weighted)													
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total
Mean Irr Req	0.00	0.00	0.00	0.38	0.61	0.19	0.00	0.00	0.00	0.00	0.00	0.00	1.18
2-in-10 Irr Req	NA	NA	NA	0.47	0.76	0.24	NA	NA	NA	NA	NA	NA	1.47
1-in-10 Irr Req	NA	NA	NA	0.52	0.83	0.26	NA	NA	NA	NA	NA	NA	1.61
Details in Million Gallons(Weighted)													
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total
Mean Irr Req	0.00	0.00	0.00	1.54	2.47	0.79	0.00	0.00	0.00	0.00	0.00	0.00	4.79
2-in-10 Irr Req	0.00	0.00	0.00	1.92	3.08	0.98	0.00	0.00	0.00	0.00	0.00	0.00	5.99
1-in-10 Irr Req	0.00	0.00	0.00	2.12	3.39	1.08	0.00	0.00	0.00	0.00	0.00	0.00	6.58

Soil Series Name : **Arents, 0 to 5 percent slopes**

Soil Map Unit Symbol# : **4** Soil Map Unit Code : **1483844** Soil Percentage : **8.15**

Details in Inches													
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total
Mean Irr Req	0.00	0.00	0.00	1.88	3.29	1.04	0.00	0.00	0.00	0.00	0.00	0.00	6.21
2-in-10 Irr Req	NA	NA	NA	2.37	4.14	1.31	NA	NA	NA	NA	NA	NA	7.82
1-in-10 Irr Req	NA	NA	NA	2.61	4.58	1.44	NA	NA	NA	NA	NA	NA	8.63
Details in Million Gallons													
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total
Mean Irr Req	0.00	0.00	0.00	7.66	13.42	4.24	0.00	0.00	0.00	0.00	0.00	0.00	25.32
2-in-10 Irr Req	0.00	0.00	0.00	9.64	16.89	5.33	0.00	0.00	0.00	0.00	0.00	0.00	31.87
1-in-10 Irr Req	0.00	0.00	0.00	10.65	18.66	5.89	0.00	0.00	0.00	0.00	0.00	0.00	35.19
Details in Inches(Weighted)													
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total
Mean Irr Req	0.00	0.00	0.00	0.15	0.27	0.08	0.00	0.00	0.00	0.00	0.00	0.00	0.51
2-in-10 Irr Req	NA	NA	NA	0.19	0.34	0.11	NA	NA	NA	NA	NA	NA	0.64
1-in-10 Irr Req	NA	NA	NA	0.21	0.37	0.12	NA	NA	NA	NA	NA	NA	0.70
Details in Million Gallons(Weighted)													
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total
Mean Irr Req	0.00	0.00	0.00	0.62	1.09	0.35	0.00	0.00	0.00	0.00	0.00	0.00	2.06
2-in-10 Irr Req	0.00	0.00	0.00	0.79	1.38	0.43	0.00	0.00	0.00	0.00	0.00	0.00	2.60
1-in-10 Irr Req	0.00	0.00	0.00	0.87	1.52	0.48	0.00	0.00	0.00	0.00	0.00	0.00	2.87

Soil Series Name : **Smyrna fine sand, 0 to 2 percent slopes**

Soil Map Unit Symbol# : **42** Soil Map Unit Code : **1483882** Soil Percentage : **56.04**

Details in Inches													
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total
Mean Irr Req	0.00	0.00	0.00	1.67	3.14	1.01	0.00	0.00	0.00	0.00	0.00	0.00	5.82
2-in-10 Irr Req	NA	NA	NA	2.11	3.97	1.27	NA	NA	NA	NA	NA	NA	7.36
1-in-10 Irr Req	NA	NA	NA	2.34	4.40	1.41	NA	NA	NA	NA	NA	NA	8.15
Details in Million Gallons													
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total
Mean Irr Req	0.00	0.00	0.00	6.83	12.82	4.10	0.00	0.00	0.00	0.00	0.00	0.00	23.74
2-in-10 Irr Req	0.00	0.00	0.00	8.62	16.19	5.18	0.00	0.00	0.00	0.00	0.00	0.00	29.99
1-in-10 Irr Req	0.00	0.00	0.00	9.54	17.94	5.74	0.00	0.00	0.00	0.00	0.00	0.00	33.22
Details in Inches(Weighted)													
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total
Mean Irr Req	0.00	0.00	0.00	0.94	1.76	0.56	0.00	0.00	0.00	0.00	0.00	0.00	3.26
2-in-10 Irr Req	NA	NA	NA	1.18	2.23	0.71	NA	NA	NA	NA	NA	NA	4.12
1-in-10 Irr Req	NA	NA	NA	1.31	2.47	0.79	NA	NA	NA	NA	NA	NA	4.57
Details in Million Gallons(Weighted)													
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total
Mean Irr Req	0.00	0.00	0.00	3.83	7.18	2.30	0.00	0.00	0.00	0.00	0.00	0.00	13.31
2-in-10 Irr Req	0.00	0.00	0.00	4.83	9.07	2.90	0.00	0.00	0.00	0.00	0.00	0.00	16.81
1-in-10 Irr Req	0.00	0.00	0.00	5.35	10.05	3.21	0.00	0.00	0.00	0.00	0.00	0.00	18.62

Soil Data:

Symbol	Soil Name	Area	Soil Type	Pct.	Soil layer info		
					SLLB	SLLL	SLDUL
42	Smyrna fine sand, 0 to 2 percent slopes	84.140	Smyrna	95%	3.9	0.062	0.130
					13.0	0.018	0.075
					20.9	0.088	0.151
					50.0	0.030	0.095
					70.1	0.020	0.080
					79.9	0.038	0.105
			Pomello	2%	3.2	0.020	0.052
					42.9	0.009	0.030
					55.1	0.027	0.100
					79.9	0.009	0.050
			Wabasso	2%	7.1	0.022	0.066
					29.1	0.014	0.060
					31.9	0.035	0.108
					38.2	0.031	0.085
					59.8	0.122	0.225
					79.9	0.035	0.091
			Placid	1%	24.0	0.047	0.188
					79.9	0.019	0.051
22	Myakka fine sand, 0 to 2 percent slopes	28.365	Myakka	90%	5.9	0.043	0.110
					20.1	0.018	0.075
					35.8	0.102	0.162
					79.9	0.023	0.084
			Basinger	5%	2.0	0.032	0.097
					18.1	0.017	0.074
					35.8	0.040	0.107
					80.3	0.019	0.077
			EauGallie	4%	5.1	0.074	0.140
					28.0	0.006	0.047
					42.1	0.064	0.131
					50.0	0.158	0.241
					65.0	0.024	0.085
			Placid	1%	3.2	0.089	0.152
					11.0	0.018	0.076
					79.9	0.024	0.085
34	Pomello fine sand, 0 to 5 percent slopes	25.402	Pomello	95%	3.9	0.032	0.104
					46.9	0.007	0.034
					57.9	0.040	0.177
					65.0	0.079	0.144
					79.9	0.017	0.074
			Smyrna	3%	7.1	0.022	0.066
					14.2	0.014	0.060
					25.2	0.035	0.108
					55.9	0.014	0.060
					79.9	0.031	0.085
			Bulow	1%	5.1	0.015	0.052
					20.1	0.012	0.044
					50.0	0.031	0.085
					59.1	0.122	0.225
					63.0	0.117	0.282
			Tavares	1%	3.2	0.015	0.052
					79.9	0.011	0.040
4	Arents, 0 to 5 percent slopes	12.242	Arents	100%	9.8	0.035	0.091
					31.9	0.035	0.091
					59.8	0.035	0.091